

ROBOT LOCALIZATION AND STATE ESTIMATION

Introduction to Robotics for AI and Data Science

Dr. Akram Fatayer



a.fatayer@zuj.edu.jo



[linkedin.com/in/akram-fatayer/](https://www.linkedin.com/in/akram-fatayer/)

Integration Map: You Are Here



Block 2: Know Where You Are

Introduction to Robotics for AI and Data Science Before your robot can plan a path, it must know where it is.

The Fundamental Problem: Where Am I?



Odometry

Measures wheel rotation.
Updates very quickly (10Hz).

Drifts over time due to cumulative error (slip, uneven terrain).



GPS

Measures absolute position.
Does not drift over time.

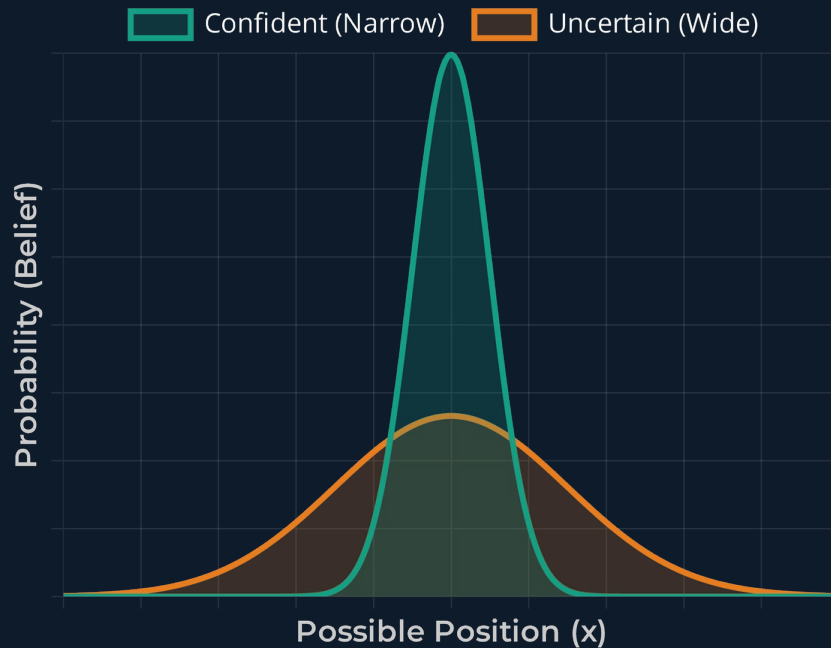
Noisy and slow (1Hz updates, random error).

Neither sensor alone gives a reliable answer.

How do we combine imperfect information to get a better estimate?

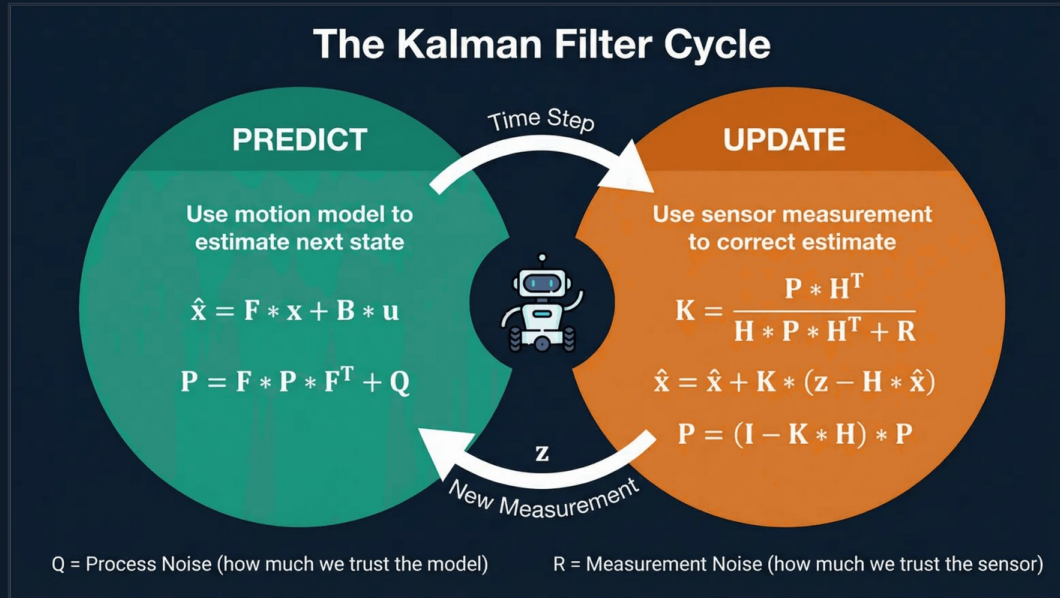
Probability as the Language of Uncertainty

- Robots rarely know their exact position — instead, they maintain a **BELIEF**.
- A belief is represented mathematically as a **probability distribution** over all possible positions.
- **Narrow distribution** = High confidence, low uncertainty. The robot is sure of its location.
- **Wide distribution** = Low confidence, high uncertainty. The robot is lost or confused.
- This paradigm shift is the foundation of modern state estimation.



— *Probabilistic Robotics* (Thrun, Burgard, Fox, 2005)

The Kalman Filter Cycle



► PREDICT

Use the robot's motion model to estimate the next state. Because models are imperfect, **uncertainty grows** during this phase.

Introduction to Robotics for AI and Data Science

🔧 UPDATE

Use new sensor measurements to correct the predicted estimate. Adding new information causes **uncertainty to shrink**.

⚖️ KALMAN GAIN (K)

The dynamic weight that decides: **Do we trust the model more, or the sensor more?** It balances Q and R noise matrices.

The Math Behind the Filter

State Vector: $x = [\text{pos_x}, \text{pos_y}, \text{vel_x}, \text{vel_y}]^T$

PREDICT

$$\hat{x} = F \cdot x + B \cdot u$$

$$P = F \cdot P \cdot F^T + Q$$

F : Transition Matrix

H : Observation Matrix

Q : Process Covariance

R : Measurement Covariance

UPDATE

$$K = P \cdot H^T \cdot (H \cdot P \cdot H^T + R)^{-1}$$

$$\hat{x} = \hat{x} + K \cdot (z - H \cdot \hat{x})$$

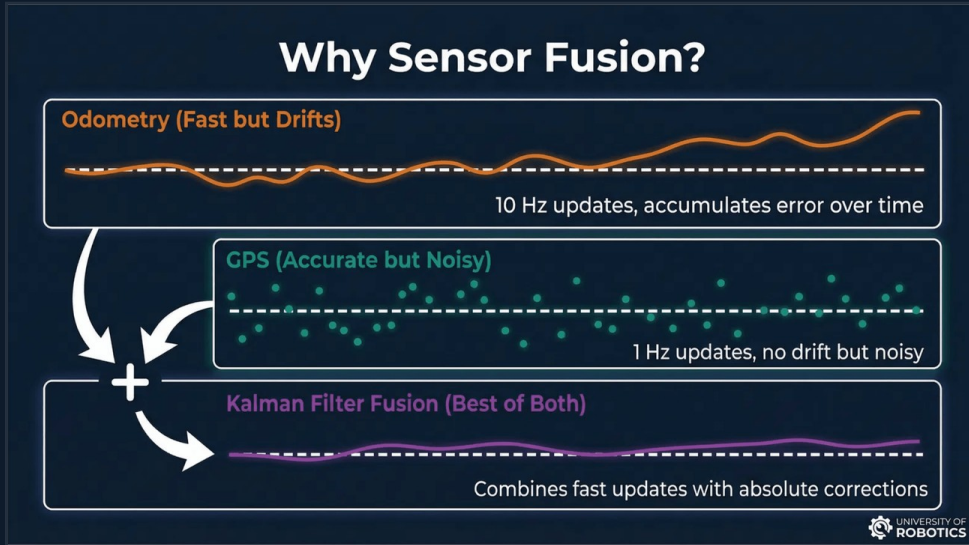
The Core Intuition

Q determines how much we trust our *motion model*.

R determines how much we trust our *sensor*.

The Kalman Gain (K) dynamically balances the two!

Why Sensor Fusion?



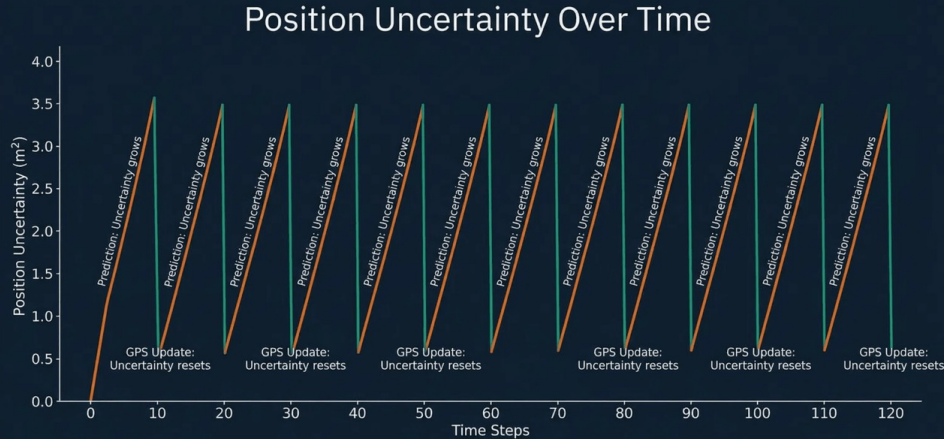
- ▶ **Odometry:** Fast (10Hz) but drifts — good for short-term estimation.
- ▶ **GPS:** Slow (1Hz) but absolute — good for long-term correction.
- ▶ **Fusion:** Combines the best of both — fast AND accurate.



Real-World Application

This is exactly what self-driving cars do, fusing data from GPS, IMU, and LiDAR to maintain precise localization.

The Uncertainty Sawtooth



This pattern proves the Kalman Filter is working: it knows when it is uncertain and corrects itself when new data arrives.



Prediction Phase

The **motion model** ($x = x + v \cdot dt$) propagates the state forward. No sensors — uncertainty **grows**.



Update Phase

Measurements correct the prediction. Odometry updates every step; GPS updates cause the **sharp drops**.



Key Insight

Both odometry and GPS are **measurements** — the prediction relies solely on the **F matrix** (motion model).



Lab Connection

You will generate and analyze this exact plot in **Activity 3**.

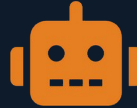
From Theory to Your Robot

This lab follows our **Incremental Integration** philosophy.
You will build your localization engine block by block.



Activity 1: Pure Math

1D Kalman Filter using NumPy.
Focus purely on the Predict/Update equations without simulation distractions.



Activity 2: The Robot

Apply the **2D Kalman Filter** to your R2D2 robot in PyBullet to track an L-shaped trajectory.



Activity 3: Sensor Fusion

Fuse noisy Odometry and GPS data to accurately track a complex circular path in the simulation.

Key Takeaways



Uncertainty is inevitable. Probability is the mathematical language we use to manage it in robotics.



No sensor is perfect. Odometry drifts over time, and GPS is noisy and slow.



The Kalman Filter optimally fuses sensors. It dynamically balances trust between the motion model and the measurements.

▶▶ **NEXT UP**

Block 3: Build Your Map

"Now that we know where we are, what does the world look like?"

Questions & Discussion

Thank you for your attention.

Dr. Akram Fatayer



a.fatayer@zuj.edu.jo



linkedin.com/in/akram-fatayer/